

Support Vector Machine (SVM) – Theory Task

Objective:

Understand the theoretical foundations of SVM, including concepts like hyperplanes, margins, support vectors, and kernel trick, and explore its real-world applications.

Task Instructions:

Answer the following questions in your own words (typed or written):

1. 1. What is a Support Vector Machine (SVM)?
 - Explain how SVM is used for classification.
 - What is the main goal of an SVM?
2. 2. What is a Hyperplane in SVM?
 - Define a hyperplane.
 - What makes a hyperplane a good separator?
3. 3. What is the Margin in SVM?
 - Define margin in the context of SVM.
 - Why does SVM try to maximize the margin?
4. 4. Who are Support Vectors?
 - What are support vectors in the SVM model?
 - How do they influence the final model?
5. 5. Linearly Separable vs. Non-Linearly Separable Data
 - What is the difference between these two types of data?
 - Can SVM work with both? How?
6. 6. What is the Kernel Trick?
 - What does a kernel do in SVM?
 - Name two commonly used kernel functions.
7. 7. Advantages and Disadvantages of SVM
 - List at least 3 advantages and 3 disadvantages.
8. 8. Real-World Applications of SVM
 - Mention 2 real-life problems where SVM is used.
 - Why is SVM suitable in those cases?
9. 🌟 Bonus (Optional):

Explain the difference between SVM and Logistic Regression in 3 points.